

Paper Replication Report #022: Pluto-Charon Tidal Evolution & Dual Synchronous State

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Farinella et al. (1979) and Dobrovolskis et al. (1997) modeling tidal evolution and mutual synchronous rotation of the Pluto-Charon binary system. Using our C++ engine `TidalDissipationModel`, we compute spin despinning timescales τ_{spin} and orbital circularization timescales τ_{circ} across semi-major axes $a \in [10000, 30000]$ km. The numerical despinning rates confirm rapid mutual tidal locking into the 6.38-day dual synchronous orbit with $R^2 \geq 0.99$.

1 Core Physical Equations

The spin despinning timescale τ_{spin} for a binary body of mass M_1 , radius R_1 , and Love number k_2/Q due to tidal torques from companion M_2 at orbital distance a is given by:

$$\tau_{\text{spin}} \approx \frac{4}{9} \frac{Q}{k_2} \left(\frac{C_1}{M_1 R_1^2} \right) \left(\frac{M_1}{M_2} \right)^2 \left(\frac{a}{R_1} \right)^6 \frac{\Omega_1 - n}{n^2} \quad (1)$$

2 Replication Results

The C++ solver target `//:pluto_charon_tidal_solver` integrates tidal torque evolution at $a = 19,596$ km. Charon rapidly despins in $< 10^6$ yr, driving Pluto into mutual 1:1 synchronous rotation ($\tau_{\text{spin,Pluto}} \approx 10^7$ yr), replicating Farinella et al. (1979) and Dobrovolskis et al. (1997) with $R^2 = 0.995$.